

Saket Deshmukh

GenAI / AI Engineer

Pune, Maharashtra — +91 7666114965 — heyiamsaket@gmail.com — linkedin.com/in/saket-deshmukh — GitHub — Portfolio

SUMMARY

GenAI Engineer with **2.5+ years of experience** designing and shipping production-grade LLM applications, specializing in **RAG pipelines, agentic workflows, hybrid retrieval, and LLM evaluation**. Skilled in building **reliable, grounded, and guardrailed AI systems** with strong observability, while optimizing retrieval quality, latency, and inference cost. Experienced across the full stack from **LLM application architecture and evaluation to cloud infrastructure, Kubernetes deployment, and CI/CD**.

SKILLS

GenAI & LLMs: RAG, Agentic AI, LLM Applications, LLM Evaluation, Prompt Engineering, Guardrails, Grounding, LLMOps

AI Frameworks: LangChain, LangGraph, OpenAI API, FastAPI, REST APIs, TruLens, LangSmith

Retrieval & Search: FAISS, BM25, pgvector, Hybrid Search, Semantic Search, Vector Search, Embeddings, RRF

Model Serving: vLLM, Self-Hosted Inference, Hugging Face

Programming: Python, SQL, Bash

Databases: PostgreSQL, MySQL

Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, Helm, Terraform, Ansible

CI/CD & Observability: GitHub Actions, Jenkins, GitLab CI, Git, Linux, Prometheus, Grafana, AWS CloudWatch

EXPERIENCE

KWAD Corporation Pvt. Ltd. — AI Engineer

Apr 2025 – Present — Pune, India

AI Compliance & Financial Operations Copilot

- Designed and built an **agentic RAG system** using **LangChain** and **LangGraph** for natural-language investigation of B2B transaction and compliance data across **12 tenants**, with conditional routing, tool calling, retrieval, response generation, and human-in-the-loop escalation.
- Engineered the **RAG pipeline** using chunking, embeddings, and **pgvector**, implementing grounded responses with source citations and reducing unsupported/hallucinated answers by an estimated **38%** based on evaluation sampling.
- Implemented **tenant-isolated retrieval** and **input/output guardrails**, replacing LLM-generated SQL with **parameterized PostgreSQL/PL-pgSQL procedures** to eliminate a class of SQL-injection and data-leak risk across **12 tenants**.
- Deployed the system on **Kubernetes** using Docker and Helm with **GitHub Actions CI/CD**; **LangSmith** tracing and RAG evaluation supported **9 iterations** of prompt/retrieval improvements with **45% faster mean-time-to-detect** regressions.

Briowin Private Limited — Software Engineer

Feb 2024 – Mar 2025 — Mumbai, India

CatalogSage – AI-Powered Product Catalog Assistant

- Built an **AI-powered RAG product assistant** for natural-language search and recommendations across a **300+ SKU catalog** using Python, FastAPI, LangGraph, and OpenAI, reducing query resolution time by an estimated **52%** versus manual search.
- Designed a **LangGraph agentic workflow** with conditional routing, input/output validation, bounded regeneration, graceful fallback, and **prompt-injection guardrails**, improving end-to-end request success rate to **94%**.
- Engineered **hybrid retrieval** using OpenAI embeddings, **FAISS**, and **BM25**, executing retrieval paths concurrently with **asyncio.gather** and merging results using **RRF**, improving top-k retrieval accuracy by **27%**; built **TruLens** evaluation with Docker/pytest CI to catch regressions in **18%** of pre-release builds.

PROJECTS

Self-Hosted LLM Inference Platform

Kubernetes, Terraform, AWS EKS, Docker, vLLM

- Built and containerized a **self-hosted LLM inference platform** using **Docker** and **vLLM**, configuring GPU acceleration, CUDA dependencies, and Python runtime to efficiently serve open-source LLMs.
- Provisioned **AWS infrastructure** including VPC, IAM, EKS, and networking using **Terraform**, and deployed the inference service on Kubernetes using Deployments, Services, and AWS Load Balancer integration.

Real-Time Product Review Summarizer

Python, NLP, TF-IDF, Random Forest, MySQL

- Built an NLP-based system to process **10,000+ product reviews** using TF-IDF vectorization and Random Forest classification for sentiment analysis, achieving approximately **90% accuracy** and reducing manual review effort by approximately **60%**.

EDUCATION

B.E. in Computer Engineering

2019 – 2023 — Pune, India

Savitribai Phule Pune University